JOSHUA S. SPEAGLE (沈佳士)

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RESEARCH INTERESTS

My research lies at the intersection of **astrophysics and cosmology**, **statistical sciences**, and **artificial intelligence**. Together with the **Astrostatistics Research Team (ART)**, which I co-lead, I develop foundational statistical methods, scientific AI and foundation models, and scalable open-source software to unlock discovery across **large sky surveys** and improve how we learn about the universe from noisy, messy data.

POSITIONS

Assistant Professor of Astrostatistics: University of Toronto — 2022-Present
Research Scientist: River Run Trading — Jan-Apr 2025
Quantitative trading strategies for crypto assets
Dunlap Postdoctoral Fellow: University of Toronto — 2020-2022
Banting Postdoctoral Fellow: University of Toronto — 2020-2022
Supervisor: Gwen Eadie
Project Academic Support Staff: Kavli IPMU, University of Tokyo — 2015-2016
Supervisors: Naoki Yoshida, Alexie Leauthaud, & Kevin Bundy

EDUCATION

Harvard University: PhD in Astronomy — 2016-2020
Advisers: Doug Finkbeiner, Charlie Conroy, & Daniel Eisenstein (with Alyssa Goodman)
Harvard University: MA in Astronomy — 2016-2020
Advisers: Daniel Eisenstein & Alexie Leauthaud
Harvard University: BA in Astrophysics and Physics — 2011-2015

SELECTED AWARDS & HONORS

Connaught New Researcher Award: University of Toronto — 2024
John Charles Polanyi Prize in Physics: Council of Ontario Universities — 2022
Best Astrostatistics Student Paper Award: Astrostatistics Interest Group (ASA) — 2020
Eric R. Keto Prize for Best Thesis in Theoretical Astrophysics: Harvard University — 2020

SELECTED GRANTS & FELLOWSHIPS

Arts & Science Collaborative Research Fund: University of Toronto — 2026-2028
CANSSI Ontario Multidisciplinary Doctoral (MDoc) Grant: CANSSI Ontario — 2026-2028
CSA JWST Cycle 2 Grant: Canadian Space Agency — 2024-2027
NSERC Discovery Grant & Launch Supplement: NSERC — 2023-2028
DSI Catalyst Grant: Data Sciences Institute, University of Toronto — 2023-2025

SUPERVISION & MENTORSHIP

I am currently (co-)supervising/mentoring **15 individuals** and have (co-)supervised/mentored **85 individuals** across Astronomy & Astrophysics, Statistical Sciences, and Computer Science, including:

9 postdoctoral researchers (8 current)
14 doctoral students (6 current)
4 master's students
57 undergraduate students (1 current) and 1 secondary school student

See my website for a full record of current group members, alumni placements, and research projects.

SELECTED LEADERSHIP & PROFESSIONAL SERVICE

Academic, Center, & Program Leadership

Astrostatistics Research Team (ART): Co-Lead (University of Toronto)	2022-Present
MScAC Data Science Concentration: Interim Director (University of Toronto)	2026-2027
Annual UofT STATSTRO Workshop: Co-Founder & Co-Organizer	2021-Present

Professional Societies & Conference Leadership

Statistical Society of Canada (SSC): Data Science & Analytics Section President / Past / Elect	2024-2027
American Astronomical Society (AAS): WGAA Steering Committee	2020-2026
LSST Informatics & Statistics Science Collaboration (ISSC): Publication Committee Chair	2023-2024
American Statistical Association (ASA): AIG Program Chair (Interim) & Web Director	2020-2024
Globular Clusters & Their Tidal Tails: SOC Co-Chair, LOC Co-Chair	2024
CASCA Annual General Meeting: SOC Co-Chair, LOC Member	2024
Canadian Astronomical Society (CASCA): Postdoc Committee	2020-2023

University & Departmental Service

Astronomy & Astrophysics: Undergraduate Curriculum Committee (2023-2024, 2025-Present), SURP Committee (Chair 2022-2024, Member 2021-2022, 2025-Present), Training & Mentoring Committee (Co-Chair 2021-2023), Graduate Admissions Committee (2020-2021)

Statistical Sciences: MScAC Data Science (Interim Director 2026-2027), Graduate Admissions Committee (2022-Present), Undergraduate Curriculum Committee (2025-Present), UTSSRP (2026)

SELECTED PRESENTATIONS & APPEARANCES

I have given **110+ presentations** related to work being done by me and my collaborators to the scientific community and the public, including **60+ invited conference talks, colloquia, and seminars**, along with providing lectures and tutorials and participating in panel discussions. See my website for a full record.

TEACHING

I have a strong interest in education, pedagogy, and curriculum development, with a focus on practical skills such as programming, applied statistics, and data science. Across Astronomy & Astrophysics and Statistical Sciences, I have developed curricula and taught **undergraduate and graduate courses** spanning first-year introductory statistical reasoning and data science, fourth-year undergraduate courses in astrophysical research and multivariate statistics, and interdisciplinary graduate courses in astrostatistics, sampling methods, and research ethics and practice, alongside lecturing at **international summer schools and workshops**. See my website for additional details on courses I have taught.

EQUITY, DIVERSITY, & INCLUSION

I am committed to improving equity, diversity, and inclusion (EDI) in the classroom, in my work, and in the wider academic community. My efforts include leading data collection and admissions reforms to expand opportunities for first-time undergraduate researchers as Chair of the **Summer Undergraduate Research Program (SURP)**, strengthening faculty, postdoc, and graduate mentoring programs and early-career industry panels as Co-Chair of the **Training & Mentoring Committee**, serving on graduate admissions and undergraduate curriculum committees across both departments, and teaching programming and data analysis for students from historically marginalized backgrounds at the **Banneker Institute**. See my website for additional details.

PUBLICATIONS

I am an author on **145+ papers** that have **19,000+ citations** (h-index=46). This includes:

- 15 papers as primary/co-first author
- 27 other papers with significant contributions (non-overlapping categories)
- 13 papers led by postdocs
- 22 papers led by students

Most of my papers can be found online on SciX, arXiv, and Google Scholar. My ORCID is 0000-0003-2573-9832. A full list of publications and additional details can also be found on my website. In the list below, individual paper citation counts are from NASA SciX as of November 2026; my name appears in **bold**, and the names of students and postdocs I have (co-)supervised appear in **bold underline**.

FIVE MOST-CITED PAPERS (NASA SciX, November 2026)

1. **Speagle, J. S.**, 2020, Monthly Notices of the Royal Astronomical Society, Vol. 493, Iss. 3, p. 3132-3158
   dynesty: A Dynamic Nested Sampling Package for Estimating Bayesian Posteriors and Evidences
   Citations: 2,669 | arXiv: 1904.02180
2. **Speagle, J. S.**; Steinhardt, C. L.; Capak, P. L.; & Silverman, J. D., 2014, The Astrophysical Journal Supplement Series, Vol. 214, Iss. 2, id. 15
   A Highly Consistent Framework for the Evolution of the Star-Forming 'Main Sequence' from z~0-6
   Citations: 1,738 | arXiv: 1405.2041
3. Green, G. M.; Schlafly, E.; Zucker, C.; **Speagle, J. S.**; & Finkbeiner, D., 2019, The Astrophysical Journal, Vol. 887, Iss. 1, id. 93
   A 3D Dust Map Based on Gaia, Pan-STARRS 1 and 2MASS
   Citations: 1,478 | arXiv: 1905.02734
4. Aihara, H. et al. [142 additional co-authors including **Speagle, J. S.**], 2018, Publications of the Astronomical Society of Japan, Vol. 70, id. S4
   The Hyper Suprime-Cam SSP Survey: Overview and Survey Design
   Citations: 1,281 | arXiv: 1702.08449
5. Johnson, B. D.; Leja, J.; Conroy, C.; & **Speagle, J. S.**, 2021, The Astrophysical Journal Supplement Series, Vol. 254, Iss. 2, id. 22
   Stellar Population Inference with Prospector
   Citations: 907 | arXiv: 2012.01426

PUBLICATIONS (FULL LIST)

(Co-)Lead Author

13 papers (refereed & under review)

13. **Zucker, C. & Saydjari, A. K. & Speagle, J. S.**; Schlafly, E. F.; Green, G. M.; Benjamin, R.; Peek, J.; Edenhofer, G.; Goodman, A.; Kuhn, M. A.; & Finkbeiner, D. P., 2025, The Astrophysical Journal, Vol. 992, Iss. 1, id. 39
    A Deep, High-Angular Resolution 3D Dust Map of the Southern Galactic Plane
    arXiv: 2503.02657

12. **Speagle, J. S.** et al. [20 additional co-authors], 2025, submitted to The Astrophysical Journal
    Deriving Stellar Properties, Distances, and Reddenings from Photometry and Astrometry with brutus
    arXiv: 2503.02227

11. **Speagle, J. S.** et al. [20 additional co-authors], 2024, The Astrophysical Journal, Vol. 970, Iss. 2, id. 121
    Mapping the Milky Way in 5-D with 170 Million Stars
    arXiv: 2503.02200

10. **Iyer, K. G. & Speagle, J. S.**; Caplar, N.; Forbes, J. C.; Gawiser, E.; Leja, J.; & Tacchella, S., 2024, The Astrophysical Journal, Vol. 961, Iss. 1, id. 53
    Stochastic Modelling of Star Formation Histories III. Constraints from Physically-Motivated Gaussian Processes
    arXiv: 2208.05938

9. **Speagle, J. S.**, 2020, Monthly Notices of the Royal Astronomical Society, Vol. 493, Iss. 3, p. 3132-3158
    dynesty: A Dynamic Nested Sampling Package for Estimating Bayesian Posteriors and Evidences
    arXiv: 1904.02180

8. **Portillo, S. K. N. & Speagle, J. S.**; & Finkbeiner, D. P., 2020, The Astronomical Journal, Vol. 159, Iss. 4, id. 165
    Photometric Biases in Modern Surveys
    arXiv: 1902.02374 **Media:** AAS

7. **Speagle, J. S.** et al. [11 additional co-authors], 2019, Monthly Notices of the Royal Astronomical Society, Vol. 490, Iss. 4, p. 5658-5677
    Galaxy-Galaxy Lensing in HSC: Validation Tests and the Impact of Heterogeneous Spectroscopic Training Sets
    arXiv: 1906.05876

6. **Zucker, C. & Speagle, J. S.**; Schlafly, E. F.; Green, G. M.; Finkbeiner, D. P.; Goodman, A. A.; & Alves, J., 2019, The Astrophysical Journal, Vol. 879, Iss. 2, id. 125
    A Large Catalog of Accurate Distances to Local Molecular Clouds: The Gaia DR2 Edition
    arXiv: 1902.01425

5. **Speagle, J. S.** & Eisenstein, D. J., 2017, Monthly Notices of the Royal Astronomical Society, Vol. 469, Iss. 1, p. 1205-1224
    Deriving Photometric Redshifts with Fuzzy Archetypes and Self-Organizing Maps II. Implementation
    arXiv: 1510.08080

4. **Speagle, J. S.** & Eisenstein, D. J., 2017, Monthly Notices of the Royal Astronomical Society, Vol. 469, Iss. 1, p. 1186-1204
   Deriving Photometric Redshifts with Fuzzy Archetypes and Self-Organizing Maps I. Methodology
   arXiv: 1510.08073

3. **Speagle, J. S.**; Capak, P. L.; Eisenstein, D. J.; Masters, D. C.; & Steinhardt, C. L., 2016, Monthly Notices of the Royal Astronomical Society, Vol. 461, Iss. 4, p. 3432-3442
   Exploring Photometric Redshifts as an Optimization Problem: An Ensemble MCMC and Simulated Annealing-Driven Template-fitting Approach
   arXiv: 1508.02484

2. **Speagle, J. S.**; Steinhardt, C. L.; Capak, P. L.; & Silverman, J. D., 2014, The Astrophysical Journal Supplement Series, Vol. 214, Iss. 2, id. 15
   A Highly Consistent Framework for the Evolution of the Star-Forming 'Main Sequence' from z~0-6
   arXiv: 1405.2041

1. **Speagle, J. S.**; Kaplan, D. L.; & van Kerkwijk, M. H., 2011, The Astrophysical Journal, Vol. 743, Iss. 2, id. 183
   The X-ray Counterpart of the High-B Pulsar J0726-2612
   arXiv: 1111.2877 **Media**: Astrobites

(Co-)Lead Author (Non-Refereed)

2 papers (invited commentaries & tutorials)

2. **Speagle, J. S. & Eadie, G. M.**, 2021, Nature Astronomy, Vol. 5, p. 971-972
   Making the Sum Greater than its Parts

1. **Speagle, J. S.**, 2019, arXiv e-prints
   A Conceptual Introduction to Markov Chain Monte Carlo Methods
   arXiv: 1909.12313

Significant Contribution

52 papers (incl. 35 student- & postdoc-led)

52. **Shen, J.**; **Speagle, J. S.**; & Ho, S., 2026, The Astrophysical Journal, Vol. 1007, Iss. 1, id. 69
    Homogeneous Stellar Parameters from Heterogeneous Spectra with Deep Learning
    arXiv: 2604.25786

51. **Heiger, M. E.**; Ji, A. P.; **Speagle, J. S.**; Li, T. S.; Savino, A.; Sandford, N. R.; Kirby, E. N.; de los Reyes, M. A. C.; & Simon, J. D., 2026, The Astrophysical Journal, Vol. 1007, Iss. 2, id. 139
    It's a Matter of Time: Empirical Constraints on Supernova Yields and Delay Times from Dwarf Spheroidal Galaxies
    arXiv: 2602.22333

50. **Xu, D.**; Stelea, I. A.; **Speagle, J. S.**; Zhang, Y.; & Tan, J. C., 2026, The Astrophysical Journal, Vol. 1001, Iss. 1, id. 120
    Disk Wind Feedback from High-mass Protostars. V. Application of Multimodal Machine Learning to Characterize Outflow Properties
    arXiv: 2601.21100

49. Hao, C.; O'Neil, S.; Vogelsberger, M.; Tran, V.; Mowla, L.; & **Speagle, J. S.**, 2026, The Open Journal of Astrophysics, Vol. 9, id. 59081
Finding the boundary: Using galaxy membership to inform galaxy cluster extent through machine learning
arXiv: 2511.07516

48. Carzon, J.; Masserano, L.; Ingram, J. D.; Shen, A.; Ribeiro Junior, A. C. H.; Dorigo, T.; Doro, M.; **Speagle, J. S.**; Izbicki, R.; & Lee, A. B., 2026, Machine Learning: Science and Technology, Vol. 7, Iss. 3, id. 035032
Trustworthy scientific inference with generative models
arXiv: 2508.02602

47. **Dey, B.** & **Speagle, J. S.**, 2026, arXiv e-prints
Tabular Foundation Models Dramatically Outperform Bespoke Photometric Redshift Estimators
arXiv: 2610.39332

46. **Savelli, A. M.**; **Kustec, M. A.**; **Speagle, J. S.**; Iyer, K. G.; **Ahmed, E.**; **Peng, Z.**; Murray, N.; & **Kuang, C.**, 2026, arXiv e-prints
Untangling Similarities and Differences in Star Formation Histories Across Cosmological Simulations
arXiv: 2610.17359

45. Géron, T.; **Laroche, A.**; **Speagle, J. S.**; & Drout, M. R., 2026, arXiv e-prints
Unresolved Binary Systems in the Rubin Era I: An Autoencoder Framework for Binary Identification Applied to 47 Tucanae
arXiv: 2609.05616

44. **Karim, T.**; **Speagle, J. S.**; & Hložek, R., 2026, arXiv e-prints
Model Predictive Scoring Shows Specific BAO Observations (not SNIa) Drives w0wa Tension
arXiv: 2609.04334

43. **Van-Lane, P. R.**; **Speagle, J. S.**; Cloutier, R.; Theissen, C. A.; Eadie, G. M.; & Kamai, I., 2026, arXiv e-prints
EncoTESS: Age-Sensitive Encodings from Raw TESS Light Curves
arXiv: 2608.25019

42. **Peng, Z.**; **Dey, B.**; Maddison, C. J.; & **Speagle, J. S.**, 2026, arXiv e-prints
Integral Field Unit Spectroscopy with One Fiber
arXiv: 2606.10197

41. Reina-Campos, M.; **Speagle, J. S.**; & Harris, W. E., 2026, arXiv e-prints
On the correlation between globular clusters and the distribution of dark matter in galaxy clusters: the case of Abell 2744
arXiv: 2605.20352

40. **Van-Lane, P. R.**; **Speagle, J. S.**; Eadie, G. M.; Douglas, S. T.; Cargile, P. A.; Zucker, C.; Lu, Y.; & Angus, R., 2025, The Astrophysical Journal, Vol. 986, Iss. 1, id. 59
ChronoFlow: A Data-Driven Model for Gyrochronology
arXiv: 2412.12244

39. **Savelli, A. M.**; **Speagle, J. S.**; Mackereth, J. T.; Murray, N.; & **Iyer, K. G.**, 2025, The Astrophysical Journal, Vol. 984, Iss. 2, id. 195
SF-R You Sure? The Conflicting Role of Star Formation Rates in Constraining the Evolution of Milky Way Analogues in Cosmological Simulations
arXiv: 2410.07244

38. **Patil, A. A.**; Eadie, G. M.; **Speagle, J. S.**; & Thomson, D. J., 2025, The Astronomical Journal, Vol. 170, Iss. 1, id. 7
Improving Harmonic Analysis using Multitapering: Precise frequency estimation of stellar oscillations using the harmonic F-test
arXiv: 2405.18509

37. **Laroche, A.** & **Speagle, J. S.**, 2025, The Astrophysical Journal, Vol. 979, Iss. 1, id. 5
Closing the stellar labels gap: Stellar label independent evidence for [α/M] information in Gaia BP/RP spectra
arXiv: 2404.07316

36. Finkbeiner, D. P.; **Speagle, J. S.**; & **Karim, T.**, 2025, arXiv e-prints
Data-Driven Stellar Spectral Modelling with GSPICE
arXiv: 2511.16754

35. **Leung, H. W.**; Bovy, J.; & **Speagle, J. S.**, 2024, ICML 2024 Workshop on Foundation Models in the Wild
Estimating Probability Densities of Tabular Data using a Transformer Model combined with Denoising Diffusion
arXiv: 2407.15703

34. **Grondin, S. M.**; Drout, M. R.; Nordhaus, J.; Muirhead, P. S.; **Speagle, J. S.**; & Chornock, R., 2024, The Astrophysical Journal, Vol. 976, Iss. 1, id. 102
The First Catalog of Candidate White Dwarf–Main-sequence Binaries in Open Star Clusters: A New Window into Common Envelope Evolution
arXiv: 2407.04775

33. **Berek, S. C.**; Eadie, G. M.; **Speagle, J. S.**; & **Wang, S. Y.**, 2024, The Astrophysical Journal, Vol. 972, Iss. 1, id. 104
Should Zeros Count? Modeling the Galaxy–Globular Cluster Scaling Relation with(out) Zero-inflated Count Models
arXiv: 2407.04764

32. Harris, W. E. & **Speagle, J. S.**, 2024, The Astronomical Journal, Vol. 168, Iss. 1, id. 38
Photometric Completeness Modelled with Neural Networks
arXiv: 2405.19135

31. **Yu, F.**; Li, T. S.; **Speagle, J. S.** et al. [9 additional co-authors], 2024, The Astrophysical Journal, Vol. 975, Iss. 1, id. 81
The Power of High Precision Broadband Photometry: Tracing the Milky Way Density Profile with Blue Horizontal Branch stars in the Dark Energy Survey
arXiv: 2402.00104

30. **Wen, R. Y.**; **Speagle, J. S.**; Webb, J. J.; & Eadie, G. M., 2024, Monthly Notices of the Royal Astronomical Society, Vol. 527, Iss. 2, p. 4193-4208
Hierarchical Bayesian inference of globular cluster properties
arXiv: 2311.03704

29. **Shen, J.**; **Speagle, J. S.**; Mackereth, J. T.; Ting, Y.-S.; & Bovy, J., 2024, The Astrophysical Journal, Vol. 960, Iss. 1, id. 84
Disentangling Stellar Age Estimates from Galactic Chemodynamical Evolution
arXiv: 2305.15634

28. **Patil, A. A.**; Eadie, G. M.; **Speagle, J. S.**; & Thomson, D. J., 2024, The Astronomical Journal, Vol. 168, Iss. 5, id. 193
Improving Power Spectral Estimation using Multitapering: Precise Asteroseismic Modeling of Stars, Exoplanets, and Beyond
arXiv: 2209.15027

27. Chen, H.; **Speagle, J. S.**; & Rogers, K. K., 2023, NeurIPS 2023 Workshop on Machine Learning and the Physical Sciences
Learning Reionization History from Quasars with Simulation-Based Inference
arXiv: 2311.16238

26. **Sun, Z.**; **Speagle, J. S.**; Huang, S.; Ting, Y.-S.; & Cai, Z., 2023, NeurIPS 2023 Workshop on Machine Learning and the Physical Sciences
Zephyr: Stitching Heterogeneous Training Data with Normalizing Flows for Photometric Redshift Inference
arXiv: 2310.20125

25. **Van-Lane, P.**; **Speagle, J. S.**; & Douglas, S., 2023, International Conference on Machine Learning Workshop on Machine Learning for Astrophysics
A Novel Application of Conditional Normalizing Flows: Stellar Age Inference with Gyrochronology
arXiv: 2307.08753

24. **Laroche, A.** & **Speagle, J. S.**, 2023, International Conference on Machine Learning Workshop on Machine Learning for Astrophysics
Closing the stellar labels gap: An unsupervised, generative model for Gaia BP/RP spectra
arXiv: 2307.06378

23. **Berek, S. C.**; Eadie, G. M.; **Speagle, J. S.**; & Harris, W. E., 2023, The Astrophysical Journal, Vol. 955, Iss. 1, id. 22
The HERBAL Model: A Hierarchical Errors-in-variables Bayesian Lognormal Hurdle Model for Galactic Globular Cluster Populations
arXiv: 2306.14945

22. Wang, B.; Leja, J.; Villar, V. A.; & **Speagle, J. S.**, 2023, The Astrophysical Journal Letters, Vol. 952, Iss. 1, id. L10
SBI++: Flexible, Ultra-fast Likelihood-free Inference Customized for Astronomical Applications
arXiv: 2304.05281

21. **Grondin, S. M.**; Webb, J. J.; Leigh, N. W. C.; **Speagle, J. S.**; & Khalifeh, R. J., 2023, Monthly Notices of the Royal Astronomical Society, Vol. 518, Iss. 3, p. 4249-4264
Searching for the Extra-Tidal Stars of Globular Clusters using High-Dimensional Analysis and a Core Particle Spray Code
arXiv: 2207.11263

20. Wang, B.; Leja, J.; Villar, A.; & **Speagle, J. S.**, 2022, Neural Information Processing System Workshop on Machine Learning and the Physical Sciences
Monte Carlo Techniques for Addressing Large Errors and Missing Data in Simulation-based Inference
arXiv: 2211.03747

19. **Tu, A. J.**; Zucker, C.; **Speagle, J. S.**; Beane, A.; Goodman, A.; Alves, J.; Faherty, J.; & Burkert, A., 2022, The Astrophysical Journal, Vol. 936, Iss. 1, id. 57
Characterizing the 3D Kinematics of Young Stars in the Radcliffe Wave
arXiv: 2208.06469

18. Zucker, C.; Goodman, A. A.; Alves, J.; Bialy, S.; Foley, M.; **Speagle, J. S.**; Großschedl, J.; Finkbeiner, D. P.; Burkert, A.; Khimey, D.; & Swiggum, C., 2022, Nature, Vol. 601, Iss. 7893, p. 334-337
Star Formation Near the Sun is Driven by Expansion of the Local Bubble
arXiv: 2201.05124 **Media:** Official Website, CBC, NYT

17. **Shen, J.**; Eadie, G. M.; Murray, N.; Zaritsky, D.; **Speagle, J. S.**; Ting, Y.-S.; Conroy, C.; Cargile, P. A.; Johnson, B. D.; Naidu, R. P.; & Han, J. J., 2022, The Astrophysical Journal, Vol. 925, Iss. 1, id. 1
The Mass of the Milky Way from the H3 Survey
arXiv: 2111.09327 **Media:** SYFY

16. Leja, J.; **Speagle, J. S.**; Ting, Y.-S.; Johnson, B. D.; Conroy, C.; Whitaker, K. E.; Nelson, E. J.; van Dokkum, P.; & Franx, M., 2022, The Astrophysical Journal, Vol. 936, Iss. 2, id. 165
A New Census of the 0.2 < z < 3.0 Universe, Part II: The Star-Forming Sequence
arXiv: 2110.04314

15. Johnson, B. D.; Leja, J.; Conroy, C.; & **Speagle, J. S.**, 2021, The Astrophysical Journal Supplement Series, Vol. 254, Iss. 2, id. 22
Stellar Population Inference with Prospector
arXiv: 2012.01426

14. **Das, K. K.**; Zucker, C.; **Speagle, J. S.**; Goodman, A.; Green, G. M.; & Alves, J., 2020, Monthly Notices of the Royal Astronomical Society, Vol. 498, Iss. 4, p. 5863-5872
Constraining the Distance to the North Polar Spur with Gaia DR2
arXiv: 2009.01320 **Media:** Quanta, CfA Science Update

13. Alves, J.; Zucker, C.; Goodman, A. A.; **Speagle, J. S.**; Meingast, S.; Robitaille, T.; Finkbeiner, D. P.; Schlafly, E. F.; & Green, G. M., 2020, Nature, Vol. 578, Iss. 7794, p. 237-239
A Galactic-scale Gas Wave in the Solar Neighborhood
arXiv: 2001.08748 **Media:** Official Website, AP, BBC

12. Zucker, C.; **Speagle, J. S.**; Schlafly, E. F.; Green, G. M.; Finkbeiner, D. P.; Goodman, A.; & Alves, J., 2020, Astronomy and Astrophysics, Vol. 633, id. A51
A Compendium of Distances to Molecular Clouds in the Star Formation Handbook
arXiv: 2001.00591

11. Leja, J.; **Speagle, J. S.**; Johnson, B. D.; Conroy, C.; van Dokkum, P.; & Franx, M., 2020, The Astrophysical Journal, Vol. 893, Iss. 2, id. 111
A New Census of the 0.2 < z < 3.0 Universe, Part I: The Stellar Mass Function
arXiv: 1910.04168

10. Cargile, P. A.; Conroy, C.; Johnson, B. D.; Ting, Y.-S.; Bonaca, A.; Dotter, A.; & **Speagle, J. S.**, 2020, The Astrophysical Journal, Vol. 900, Iss. 1, id. 28
MINESweeper: Spectrophotometric Modeling of Stars in the Gaia Era
arXiv: 1907.07690

9. Safarzadeh, M.; Berger, E.; Leja, J.; & **Speagle, J. S.**, 2019, The Astrophysical Journal Letters, Vol. 878, Iss. 1, id. L14
Measuring the Delay Time Distribution of Binary Neutron Stars III. Using the Individual Star Formation Histories of Gravitational Wave Event Host Galaxies in the Local Universe
arXiv: 1905.04310 Media: AAS NOVA

8. Green, G. M.; Schlafly, E.; Zucker, C.; **Speagle, J. S.**; & Finkbeiner, D., 2019, The Astrophysical Journal, Vol. 887, Iss. 1, id. 93
A 3D Dust Map Based on Gaia, Pan-STARRS 1 and 2MASS
arXiv: 1905.02734

7. Cook, B. A.; Conroy, C.; van Dokkum, P.; & **Speagle, J. S.**, 2019, The Astrophysical Journal, Vol. 876, Iss. 1, id. 78
Measuring Star-Formation Histories, Distances, and Metallicities with Pixel Color-Magnitude Diagrams I: Model Definition and Mock Tests
arXiv: 1904.00011

6. Leja, J.; Carnall, A. C.; Johnson, B. D.; Conroy, C.; & **Speagle, J. S.**, 2019, The Astrophysical Journal, Vol. 876, Iss. 1, id. 3
How to Measure Galaxy Star Formation Histories II: Nonparametric Models
arXiv: 1811.03637

5. Zucker, C.; Schlafly, E. F.; **Speagle, J. S.**; Green, G. M.; Portillo, S. K. N.; Finkbeiner, D. P.; & Goodman, A. A., 2018, The Astrophysical Journal, Vol. 869, Iss. 1, id. 83
Mapping Distances Across the Perseus Molecular Cloud Using CO Observations, Stellar Photometry, and Gaia DR2 Parallax Measurements
arXiv: 1803.08931

4. Tanaka, M.; Coupon, J.; Hsieh, B.-C.; Mineo, S.; Nishizawa, A. J.; **Speagle, J. S.**; Furusawa, H.; Miyazaki, S.; & Murayama, H., 2018, Publications of the Astronomical Society of Japan, Vol. 70, id. S9
Photometric Redshifts for the Hyper Suprime-Cam Subaru Strategic Program Data Release 1
arXiv: 1704.05988

3. Steinhardt, C. L.; Capak, P.; Masters, D.; & **Speagle, J. S.**, 2016, The Astrophysical Journal, Vol. 824, Iss. 1, id. 21
The Impossibly Early Galaxy Problem
arXiv: 1506.01377

2. Steinhardt, C. L. & **Speagle, J. S.**, 2014, The Astrophysical Journal, Vol. 796, Iss. 1, id. 25
A Uniform History for Galaxy Evolution
arXiv: 1409.2883

1. Steinhardt, C. L.; **Speagle, J. S.** et al. [22 additional co-authors], 2014, The Astrophysical Journal Letters, Vol. 791, Iss. 2, id. L25
Star Formation at $4 < z < 6$ from the Spitzer Large Area Survey with Hyper-Suprime-Cam (SPLASH)
arXiv: 1407.7030 Media: JPL

67. **Kerr, R.**; Kraus, A. L.; Tan, J. C.; Chanamé, J.; Pérez Paolino, F.; **Speagle, J. S.**; Farias, J. P.; Fernández-Trincado, J. G.; & Hawkins, K., 2026, The Astrophysical Journal, Vol. 1004, Iss. 1, id. 132
SPYGLASS. VII-B. Tracing the Fragments of Massive Star Formation Using Low-mass Associations
arXiv: 2604.27067

66. **Xu, D.**; Martin, P. G.; Offner, S. S. R.; Gutermuth, R.; Grudić, M. Y.; & **Speagle, J. S.**, 2026, The Astrophysical Journal Supplement Series, Vol. 283, Iss. 2, id. 67
Assessing Zeeman Measurements of Magnetic Fields in Synthetic H I Observations
arXiv: 2602.18713

65. Burnham, E. et al. [15 additional co-authors including **Speagle, J. S.**], 2026, The Astrophysical Journal, Vol. 1001, Iss. 2, id. 205
It's More Complicated Than You Think: A Forward Model to Infer the Recent Star Formation History, Bursty or Not, of Galaxy Populations
arXiv: 2601.20930

64. Habib, I. et al. [13 additional co-authors including **Speagle, J. S.**], 2026, arXiv e-prints
The Great Chat in the Sky: Aligning Exotic Astronomical Data to Language with AstroBridge
arXiv: 2610.36375

63. Sarkar, M.; Ghosh, A.; Connolly, A.; Urry, C. M.; Coli, A.; **Speagle, J. S.**; Shimakawa, R.; Willard, C.; & Lockyer, D., 2026, arXiv e-prints
Complete Multi-Band Structural Characterization for 10 Million Galaxies in the Hyper Suprime-Cam Wide Survey using Deep Learning
arXiv: 2610.32913

62. Jaiswar, R. et al. [25 additional co-authors including **Speagle, J. S.**], 2026, *Publications of the Astronomical Society of Australia*, Vol. 43, id. e117
OutThere Survey: Addressing ξ_ion and f_esc with a population of average galaxies at z~2
arXiv: 2608.19687

61. **Poon, M.**; Bryan, M. L.; Rein, H.; Dong, J.; **Speagle, J. S.**; & Pham, D., 2025, The Astrophysical Journal Letters, Vol. 994, Iss. 2, id. L48
Early Evidence for Isotropic Planetary Obliquities in Young Super-Jupiter Systems
arXiv: 2511.04091

60. **Kerr, R.**; Pérez Paolino, F.; Tan, J. C.; **Speagle, J. S.**; Kraus, A. L.; Fernández-Trincado, J. G.; Stassun, K. G.; & Chanamé, J., 2025, The Astrophysical Journal, Vol. 995, Iss. 2, id. 217
SPYGLASS. VII-A. The Demographics and Ages of Small Nearby Young Associations
arXiv: 2510.21936

59. Iyer, K. G. et al. [21 additional co-authors including **Speagle, J. S.**], 2025, The Astrophysical Journal, Vol. 994, Iss. 2, id. 174
How Does Feedback Affect the Star Formation Histories of Galaxies?
arXiv: 2508.21152

58. **Kerr, R.**; Farias, J. P.; Prato, L.; Rector, T. A.; **Speagle, J. S.**; & Kraus, A. L., 2025, The Astrophysical Journal, Vol. 985, Iss. 1, id. 111
SPYGLASS. VI. Feedback-driven Star Formation in the Circinus Complex
arXiv: 2503.02002

57. Li, D.; Eadie, G. M.; Brown, P. E.; Harris, W. E.; Abraham, R. G.; van Dokkum, P.; Janssens, S. R.; **Berek, S. C.**; Danieli, S.; Romanowsky, A. J.; & **Speagle, J. S.**, 2025, The Astrophysical Journal, Vol. 984, Iss. 2, id. 147
Discovery of Two Ultra-Diffuse Galaxies with Unusually Bright Globular Cluster Luminosity Functions via a Mark-Dependently Thinned Point Process (MATHPOP)
arXiv: 2409.06040

56. **Karim, T.** et al. [40 additional co-authors including **Speagle, J. S.**], 2025, Journal of Cosmology and Astroparticle Physics, Vol. 2025, Iss. 2, id. 045
Measuring σ8 using DESI Legacy Imaging Surveys Emission-Line Galaxies and Planck CMB Lensing and the Impact of Dust on Parameter Inference
arXiv: 2408.15909

55. Esquivel, J. A.; **Shen, Y.**; Leos-Barajas, V.; Eadie, G.; **Speagle, J. S.**; Craiu, R. V.; Medina, A.; & Davenport, J. R. A., 2025, The Astrophysical Journal, Vol. 979, Iss. 2, id. 141
Detecting stellar flares in photometric data using hidden Markov models
arXiv: 2404.13145

54. **Antwi-Danso, J.** et al. [14 additional co-authors including **Speagle, J. S.**], 2025, The Astrophysical Journal, Vol. 978, Iss. 1, id. 90
The FENIKS Survey: Spectroscopic Confirmation of Massive Quiescent Galaxies at z ~ 3–5
arXiv: 2307.09590

53. Walmsley, M. et al. [18 additional co-authors including **Speagle, J. S.**], 2025, arXiv e-prints
Galaxy Zoo Evo: 1 million human-annotated images of galaxies
arXiv: 2512.23691

52. **Antwi-Danso, J.** et al. [19 additional co-authors including **Speagle, J. S.**], 2025, arXiv e-prints
An Ancient Descendant of the First Galaxies
arXiv: 2512.03154

51. Muzzin, A. et al. [82 additional co-authors including **Speagle, J. S.**], 2025, arXiv e-prints
MINERVA: A NIRCam Medium Band and MIRI Imaging Survey to Unlock the Hidden Gems of the Distant Universe
arXiv: 2507.19706

50. Wan, J. T.; Tacchella, S.; Johnson, B. D.; Iyer, K. G.; **Speagle, J. S.**; & Maiolino, R., 2024, Monthly Notices of the Royal Astronomical Society, Vol. 532, Iss. 4, p. 4002-4025
Stochastic prior for non-parametric star-formation histories
arXiv: 2404.14494

49. Alfonzo, J. P.; Iyer, K. G.; Akiyama, M.; Bryan, G. L.; Cooray, S.; Ludwig, E.; Mowla, L.; Omori, K. C.; Pacifici, C.; **Speagle, J. S.**; & Wu, J. F., 2024, The Astrophysical Journal, Vol. 967, Iss. 2, id. 152
Katachi (形): Decoding the Imprints of Past Star Formation on Present-day Morphology in Galaxies with Interpretable CNNs
arXiv: 2404.05146

48. **Grondin, S. M.**; Webb, J. J.; Lane, J. M. M.; **Speagle, J. S.**; & Leigh, N. W. C., 2024, Monthly Notices of the Royal Astronomical Society, Vol. 528, Iss. 3, p. 5189-5211
A catalogue of Galactic GEMS: Globular cluster Extra-tidal Mock Stars
arXiv: 2310.09331

47. McCullough, J. et al. [48 additional co-authors including **Speagle, J. S.**], 2024, Monthly Notices of the Royal Astronomical Society, Vol. 531, Iss. 2, p. 2582-2602
DESI Complete Calibration of the Color-Redshift Relation (DC3R2): Results from early DESI data
arXiv: 2309.13109

46. Limberg, G. et al. [11 additional co-authors including **Speagle, J. S.**], 2024, Monthly Notices of the Royal Astronomical Society, Vol. 530, Iss. 3, p. 2512-2525
Extending the Chemical Reach of the H3 Survey: Detailed Abundances of the Dwarf-galaxy Stellar Stream Wukong/LMS-1
arXiv: 2308.13702

45. Hwang, H.-C.; Ting, Y.-S.; Cheng, S.; & **Speagle, J. S.**, 2024, Monthly Notices of the Royal Astronomical Society, Vol. 528, Iss. 3, p. 4272-4288
Dynamical masses across the Hertzsprung-Russell diagram
arXiv: 2308.08584

44. Edenhofer, G.; Zucker, C.; Frank, P.; Saydjari, A. K.; **Speagle, J. S.**; Finkbeiner, D.; & Enßlin, T. A., 2024, Astronomy and Astrophysics, Vol. 685, id. A82
A Parsec-Scale Galactic 3D Dust Map out to 1.25 kpc from the Sun
arXiv: 2308.01295

43. Walmsley, M. et al. [18 additional co-authors including **Speagle, J. S.**], 2024, arXiv e-prints
Scaling Laws for Galaxy Images
arXiv: 2404.02973

42. Mathews, E. P.; Leja, J.; **Speagle, J. S.**; Johnson, B. D.; Gibson, J.; Nelson, E. J.; Suess, K. A.; Tacchella, S.; Whitaker, K. E.; & Wang, B., 2023, The Astrophysical Journal, Vol. 954, Iss. 2, id. 132
As Simple as Possible but No Simpler: Optimizing the Performance of Neural Net Emulators for Galaxy SED Fitting
arXiv: 2306.16442

41. O'Grady, A. J. G. et al. [10 additional co-authors including **Speagle, J. S.**], 2023, The Astrophysical Journal, Vol. 943, Iss. 1, id. 18
Cool, Luminous, and Highly Variable Stars in the Magellanic Clouds. II. Spectroscopic and Environmental Analysis of Thorne-Żytkow Object and Super-AGB Star Candidates
arXiv: 2211.12438

40. Johnson, J. W.; Conroy, C.; Johnson, B. D.; Peter, A. H. G.; Cargile, P. A.; Bonaca, A.; Naidu, R. P.; Woody, T.; Ting, Y.-S.; Han, J. J.; & **Speagle, J. S.**, 2023, Monthly Notices of the Royal Astronomical Society, Vol. 526, Iss. 4, p. 5084-5109
Dwarf galaxy archaeology from chemical abundances and star formation histories
arXiv: 2210.01816

39. Nelson, E. J. et al. [24 additional co-authors including **Speagle, J. S.**], 2023, The Astrophysical Journal Letters, Vol. 948, Iss. 2, id. L18
JWST Reveals a Population of Ultrared, Flattened Galaxies at $2 \lesssim z \lesssim 6$ Previously Missed by HST
arXiv: 2208.01630

38. Saydjari, A. K. et al. [11 additional co-authors including **Speagle, J. S.**], 2023, The Astrophysical Journal Supplement Series, Vol. 264, Iss. 2, id. 28
The Dark Energy Camera Plane Survey 2 (DECaPS2): More Sky, Less Bias, and Better Uncertainties
arXiv: 2206.11909

37. Han, J. J.; Conroy, C.; Johnson, B. D.; **Speagle, J. S.**; Bonaca, A.; Chandra, V.; Naidu, R. P.; Ting, Y.-S.; Woody, R.; & Zaritsky, D., 2022, The Astronomical Journal, Vol. 164, Iss. 6, id. 249
The Stellar Halo of the Galaxy is Tilted and Doubly Broken
arXiv: 2208.04327

36. Chandra, V. et al. [10 additional co-authors including **Speagle, J. S.**], 2022, The Astrophysical Journal, Vol. 940, Iss. 2, id. 127
A Ghost in Boötes: The Least Luminous Disrupted Dwarf Galaxy
arXiv: 2207.13717

35. Ashton, G. et al. [21 additional co-authors including **Speagle, J. S.**], 2022, Nature Reviews Methods Primers, Vol. 2, id. 39
Nested Sampling for Physical Scientists
arXiv: 2205.15570

34. Han, J. J. et al. [10 additional co-authors including **Speagle, J. S.**], 2022, The Astrophysical Journal, Vol. 934, Iss. 1, id. 14
A Tilt in the Dark Matter Halo of the Galaxy
arXiv: 2202.07662

33. Emami, R.; Hernquist, L.; Vogelsberger, M.; Shen, X.; **Speagle, J. S.**; Moreno, J.; Alcock, C.; Genel, S.; Forbes, J. C.; Marinacci, F.; & Torrey, P., 2022, The Astrophysical Journal, Vol. 937, Iss. 1, id. 20
On the Robustness of the Velocity Anisotropy Parameter in Probing the Stellar Kinematics in Milky Way-like Galaxies: Takeaways from the TNG50 Simulation
arXiv: 2202.07162

32. Leauthaud, A. et al. [105 additional co-authors including **Speagle, J. S.**], 2022, Monthly Notices of the Royal Astronomical Society, Vol. 510, Iss. 4, p. 6150-6189
Lensing Without Borders – I. A Blind Comparison of the Amplitude of Galaxy-Galaxy Lensing Between Independent Imaging Surveys
arXiv: 2111.13805

31. Sugiyama, S. et al. [25 additional co-authors including **Speagle, J. S.**], 2022, Physical Review D, Vol. 105, Iss. 12, id. 123537
HSC Year 1 Cosmology Results with the Minimal Bias Method: HSC x BOSS Galaxy-Galaxy Weak Lensing and BOSS Galaxy Clustering
arXiv: 2111.10966

30. Miyatake, H. et al. [27 additional co-authors including **Speagle, J. S.**], 2022, Physical Review D, Vol. 106, Iss. 8, id. 083520
Cosmological Inference from the Emulator Based Halo Model II: Joint Analysis of Galaxy-Galaxy Weak Lensing and Galaxy Clustering from HSC-Y1 and SDSS
arXiv: 2111.02419

29. Hwang, H.-C.; Ting, Y.-S.; Conroy, C.; Zakamska, N. L.; El-Badry, K.; Cargile, P.; Zaritsky, D.; Chandra, V.; Han, J. J.; **Speagle, J. S.**; & Bonaca, A., 2022, Monthly Notices of the Royal Astronomical Society, Vol. 513, Iss. 1, p. 754-767
Wide Binaries from the H3 Survey: The Thick Disc and Halo have Similar Wide Binary Fractions
arXiv: 2111.01788

28. Naidu, R. P. et al. [11 additional co-authors including **Speagle, J. S.**], 2022, The Astrophysical Journal Letters, Vol. 926, Iss. 2, id. L36
Evidence from Disrupted Halo Dwarfs that r-process Enrichment via Neutron Star Mergers is Delayed by > 500 Myr
arXiv: 2110.14652

27. Huang, S.; Leauthaud, A.; Bradshaw, C.; Hearin, A.; Behroozi, P.; Lange, J.; Greene, J.; DeRose, J.; **Speagle, J. S.**; & Xhakaj, E., 2022, Monthly Notices of the Royal Astronomical Society, Vol. 515, Iss. 4, p. 4722-4752
The Outer Stellar Mass of Massive Galaxies: A Simple Tracer of Halo Mass with Scatter Comparable to Richness and Reduced Projection Effects
arXiv: 2109.02646

26. Tacchella, S. et al. [15 additional co-authors including **Speagle, J. S.**], 2022, The Astrophysical Journal, Vol. 926, Iss. 2, id. 134
Fast, Slow, Early, Late: Quenching Massive Galaxies at z ~ 0.8
arXiv: 2102.12494

25. Naidu, R. P. et al. [10 additional co-authors including **Speagle, J. S.**], 2022, submitted to The Astrophysical Journal
Live Fast, Die α-Enhanced: The Mass-Metallicity-α Relation of the Milky Way's Disrupted Dwarf Galaxies
arXiv: 2204.09057

24. Conroy, C. et al. [13 additional co-authors including **Speagle, J. S.**], 2022, arXiv e-prints
Birth of the Galactic Disk Revealed by the H3 Survey
arXiv: 2204.02989

23. Zucker, C. et al. [10 additional co-authors including **Speagle, J. S.**], 2021, The Astrophysical Journal, Vol. 919, Iss. 1, id. 35
On the Three-dimensional Structure of Local Molecular Clouds
arXiv: 2109.09765

22. Naidu, R. P.; Conroy, C.; Bonaca, A.; Zaritsky, D.; Weinberger, R.; Ting, Y.-S.; Caldwell, N.; Tacchella, S.; Han, J. J.; **Speagle, J. S.**; & Cargile, P. A., 2021, The Astrophysical Journal, Vol. 923, Iss. 1, id. 92
Reconstructing the Last Major Merger of the Milky Way with the H3 Survey
arXiv: 2103.03251

21. Nelson, E. J. et al. [23 additional co-authors including **Speagle, J. S.**], 2021, Monthly Notices of the Royal Astronomical Society, Vol. 508, Iss. 1, p. 219-235
Spatially Resolved Star Formation and Inside-Out Quenching in the TNG50 Simulation and 3D-HST Observations
arXiv: 2101.12212

20. Emami, R. et al. [10 additional co-authors including **Speagle, J. S.**], 2021, The Astrophysical Journal, Vol. 918, Iss. 1, id. 7
Inferring the Morphology of Stellar Distributions in TNG50: Twisted and Twisted-Stretched Shapes
arXiv: 2012.12284

19. Bonaca, A. et al. [10 additional co-authors including **Speagle, J. S.**], 2021, The Astrophysical Journal Letters, Vol. 909, Iss. 2, id. L26
Orbital Clustering Identifies the Origins of Galactic Stellar Streams
arXiv: 2012.09171

18. Carter, C.; Conroy, C.; Zaritsky, D.; Ting, Y.-S.; Bonaca, A.; Naidu, R. P.; Johnson, B. D.; Cargile, P. A.; Caldwell, N.; **Speagle, J. S.**; & Han, J. J., 2021, The Astrophysical Journal, Vol. 908, Iss. 2, id. 208
Ancient Very Metal-Poor Stars Associated with the Galactic Disk in the H3 Survey
arXiv: 2012.00036

17. Green, G. M.; Rix, H.-W.; Tschesche, L.; Finkbeiner, D.; Zucker, C.; Schlafly, E. F.; Rybizki, J.; Fouesneau, M.; Andrae, R.; & **Speagle, J. S.**, 2021, The Astrophysical Journal, Vol. 907, Iss. 1, id. 57
Data-Driven Stellar Models
arXiv: 2006.16258

16. Zaritsky, D. et al. [10 additional co-authors including **Speagle, J. S.**], 2020, The Astrophysical Journal Letters, Vol. 905, Iss. 1, id. L3
Discovery of Magellanic Stellar Debris in the H3 Survey
arXiv: 2011.09395

15. Euclid Collaboration et al. [171 additional co-authors including **Speagle, J. S.**], 2020, Astronomy and Astrophysics, Vol. 644, id. A31
Euclid Preparation. X. The Euclid Photometric-Redshift Challenge
arXiv: 2009.12112

14. Johnson, B. D.; Conroy, C.; Naidu, R. P.; Bonaca, A.; Zaritsky, D.; Ting, Y.-S.; Cargile, P. A.; Han, J. J.; & **Speagle, J. S.**, 2020, The Astrophysical Journal, Vol. 900, Iss. 2, id. 103
A Diffuse Metal-Poor Component of the Sagittarius Stream Revealed by the H3 Survey
arXiv: 2007.14408

13. Cabrera-Ziri, I.; **Speagle, J. S.** et al. [10 additional co-authors], 2020, Monthly Notices of the Royal Astronomical Society, Vol. 495, Iss. 1, p. 375-382
Searching for Globular Cluster Chemical Anomalies on the Main Sequence of a Young Massive Cluster
arXiv: 2004.09636

12. Bonaca, A.; Conroy, C.; Hogg, D. W.; Cargile, P. A.; Caldwell, N.; Naidu, R. P.; Price-Whelan, A. M.; **Speagle, J. S.**; & Johnson, B. D., 2020, The Astrophysical Journal Letters, Vol. 892, Iss. 2, id. L37

High-Resolution Spectroscopy of the GD-1 Stellar Stream Localizes the Perturber Near the Orbital Plane of Sagittarius

arXiv: 2001.07215

11. Huang, S. et al. [10 additional co-authors including **Speagle, J. S.**], 2020, Monthly Notices of the Royal Astronomical Society, Vol. 492, Iss. 3, p. 3685-3707

Weak Lensing Reveals a Tight Connection Between Dark Matter Halo Mass and the Distribution of Stellar Mass in Massive Galaxies

arXiv: 1811.01139 **Media:** CfA Science Update

10. Namikawa, T. et al. [73 additional co-authors including **Speagle, J. S.**], 2019, The Astrophysical Journal, Vol. 882, Iss. 1, id. 62

Evidence for the Cross-correlation between Cosmic Microwave Background Polarization Lensing from POLARBEAR and the Cosmic Shear from Subaru Hyper Suprime-Cam

arXiv: 1904.02116

9. Leja, J.; Johnson, B. D.; Conroy, C.; van Dokkum, P.; **Speagle, J. S.**; Brammer, G.; Momcheva, I.; Skelton, R.; Whitaker, K. E.; Franx, M.; & Nelson, E. J., 2019, The Astrophysical Journal, Vol. 877, Iss. 2, id. 140

An Older, More Quiescent Universe from Panchromatic SED Fitting of the 3D-HST Survey

arXiv: 1812.05608

8. Forbes, J. C.; Krumholz, M. R.; & **Speagle, J. S.**, 2019, Monthly Notices of the Royal Astronomical Society, Vol. 487, Iss. 3, p. 3581-3606

Towards a Radially-Resolved Semi-Analytic Model for the Evolution of Disc Galaxies Tuned with Machine Learning

arXiv: 1810.12919

7. Hikage, C. et al. [35 additional co-authors including **Speagle, J. S.**], 2019, Publications of the Astronomical Society of Japan, Vol. 71, Iss. 2, id. 43

Cosmology from cosmic shear power spectra with Subaru Hyper Suprime-Cam first-year data

arXiv: 1809.09148 **Media:** PASJ Excellent Paper Award (English)

6. Medezinski, E. et al. [14 additional co-authors including **Speagle, J. S.**], 2018, Publications of the Astronomical Society of Japan, Vol. 70, Iss. 2, id. 30

Source Selection for Cluster Weak Lensing Measurements in the Hyper Sprime-Cam Survey

arXiv: 1706.00427

5. Mandelbaum, R. et al. [29 additional co-authors including **Speagle, J. S.**], 2018, Publications of the Astronomical Society of Japan, Vol. 70, id. S25

The first-year shear catalog of the Subaru Hyper Suprime-Cam SSP Survey

arXiv: 1705.06745

4. Aihara, H. et al. [142 additional co-authors including **Speagle, J. S.**], 2018, Publications of the Astronomical Society of Japan, Vol. 70, id. S4

The Hyper Suprime-Cam SSP Survey: Overview and Survey Design

arXiv: 1704.05858

3. Aihara, H. et al. [108 additional co-authors including **Speagle, J. S.**], 2018, Publications of the Astronomical Society of Japan, Vol. 70, id. S8
   First Data Release of the Hyper Suprime-Cam Subaru Strategic Program
   arXiv: 1702.08449

2. Oguri, M. et al. [23 additional co-authors including **Speagle, J. S.**], 2018, Publications of the Astronomical Society of Japan, Vol. 70, id. S20
   An optically-selected cluster catalog at redshift 0.1<z<1.1 from Hyper Suprime-Cam Subaru Strategic Program S16A data
   arXiv: 1701.00818

1. Masters, D. et al. [18 additional co-authors including **Speagle, J. S.**], 2015, The Astrophysical Journal, Vol. 813, Iss. 1, id. 53
   Mapping the Galaxy Color-Redshift Relation: Optimal Photometric Redshift Calibration Strategies for Cosmology Surveys
   arXiv: 1509.03318

Contributing Author (Non-Refereed)

7 papers (white papers & community primers)

7. Ntampaka, M. et al. [15 additional co-authors including **Speagle, J. S.**], 2026, arXiv e-prints
   How to Craft the Right Language AI Policy For Your Research Group (Some Assembly Required)
   arXiv: 2607.20836

6. Paladini, R. et al. [68 additional co-authors including **Speagle, J. S.**], 2023, arXiv e-prints
   Roman Early-Definition Astrophysics Survey Opportunity: Galactic Roman Infrared Plane Survey (GRIPS)
   arXiv: 2307.07642

5. Han, J. J. et al. [207 additional co-authors including **Speagle, J. S.**], 2023, arXiv e-prints
   NANCY: Next-generation All-sky Near-infrared Community surveY
   arXiv: 2306.11784

4. Eadie, G. M.; **Speagle, J. S.**; Cisewski-Kehe, J.; Foreman-Mackey, D.; Huppenkothen, D.; Jones, D. E.; Springford, A.; & Tak, H., 2023, arXiv e-prints
   Practical Guidance for Bayesian Inference in Astronomy
   arXiv: 2302.04703

3. Ntampaka, M.; Bonaca, A.; Bose, S.; Eisenstein, D. J.; Hadzhiyska, B.; Mason, C.; Nagai, D.; & **Speagle, J. S.**, 2022, Bulletin of the American Astronomical Society, Vol. 54, Iss. 1, id. 51
   A Referee Primer for Early Career Astronomers
   arXiv: 2205.14270

2. Siemiginowska, A. et al. [50 additional co-authors including **Speagle, J. S.**], 2019, Bulletin of the American Astronomical Society, Vol. 51, Iss. 3, id. 355
   The Next Decade of Astroinformatics and Astrostatistics
   arXiv: 1903.06796

1. Zasowski, G. et al. [10 additional co-authors including **Speagle, J. S.**], 2019, Bulletin of the American Astronomical Society, Vol. 51, Iss. 3, id. 314
   High-Dimensional Dust Mapping
   arXiv: 1903.05150